5 Installation and Dismantling RR 300 plus

5.1 Inspecting the delivery

For the most part the RR 300 plus is pre-assembled when delivered. The side columns are fixed together with the top roll cover, the motor cover (if included in delivery), the fixing material and the control unit, to a transport pallet (made of wood). Check the door for possible transport damage, and ensure that the shipment is complete by comparing the order documents.

5.2 Installation preparation

Inspect the install location relative to the required anchoring points and set-up conditions. It may be necessary to prepare a support construction or other suitable anchoring possibilities. The head plates should be bolted to a steel support construction or to a concrete element.

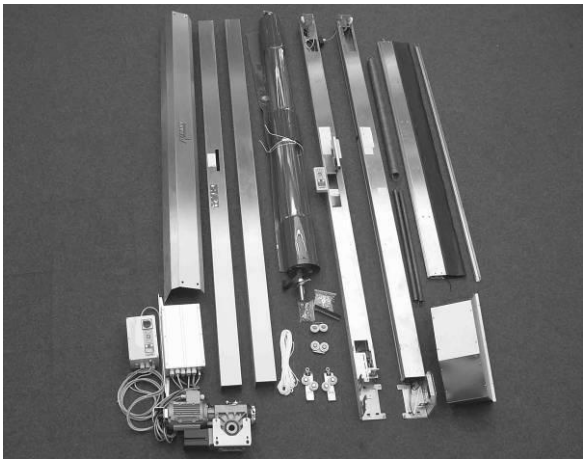


Check isolated panel walls, rider brick, and lightweight construction walls to ensure that they are suitable for the installation.

Measure the clear dimensions (width and height) of the door opening and compare these dimensions with the order information.

The install location must be safeguarded from pedestrian and vehicle traffic.

Remove the packaging from the door prior to installation. Remove the covers of the side frames and the control unit from the pallet and place them a good distance from the door opening.

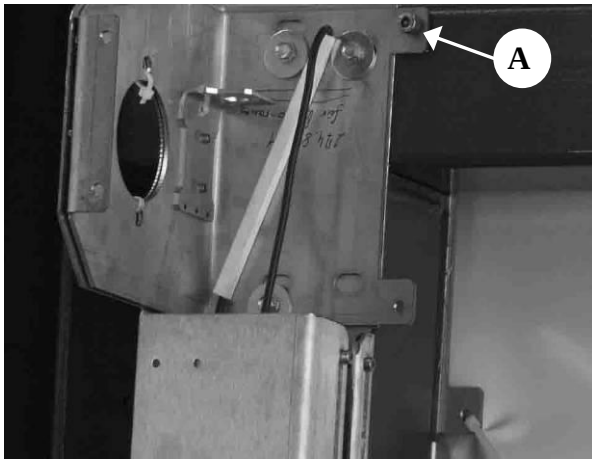




1. Check the door for possible transport damage, and ensure that the delivery is complete by comparing the shipping documents.

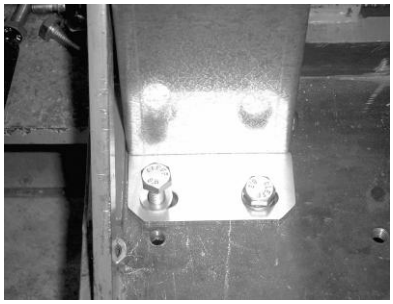
2. The drive unit and the installation material are shipped together in one cardboard box.





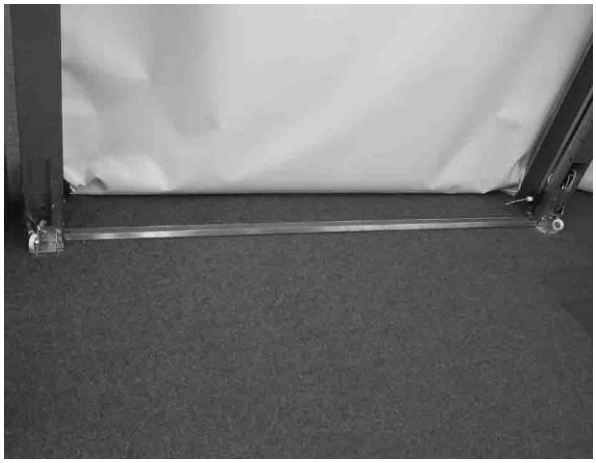
3. If the floor is inclined then the first side frame must be installed on the higher side. Align the first side frame so that it is precisely vertical; you can use e.g. a water level.

4. Anchor the head plate to the wall with 3 bolts.
Position **A** shows the bore for anchoring the cable conduit (without option sealed). Mount the cable conduit later.

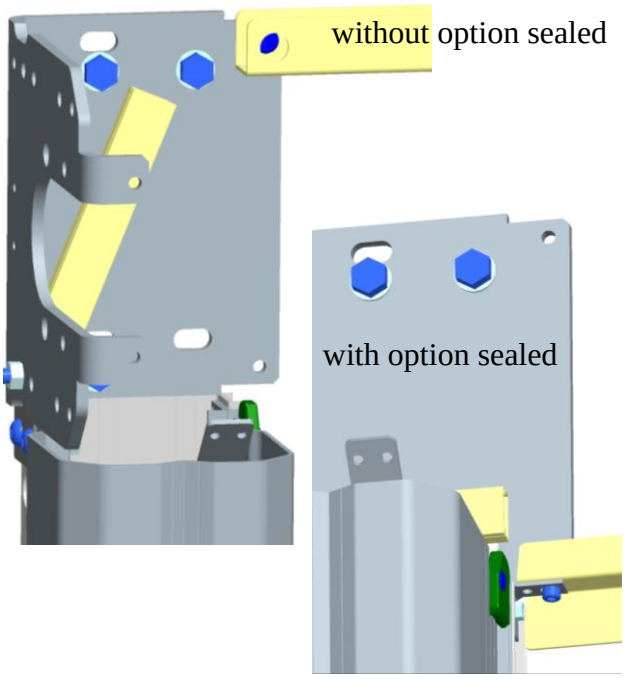




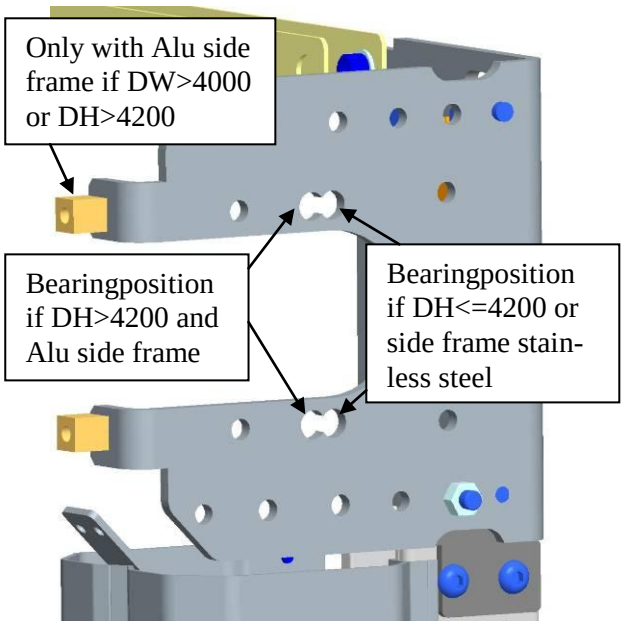
5. Fix the floor plate in place with 2 bolts. Alternative the side frame can be fixed at the wall with 1 bolt.



6. Check the length of the cable conduit, must be DW. Set up the second side frame with the aid of the cable conduit (=exact clear width). The height differential between the side frames is max. 3 mm. Shim the lower side if necessary.



7. Screw the cable conduit to the head plates at the points provided.



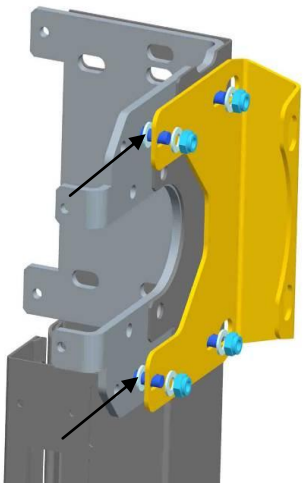
8. Use a crane, forklift, or similar lift gear, to raise the top roll and anchor it on the head plates. If the door is small, the top roll can be handled by hand (2 persons). See also pictures 9 and 10.



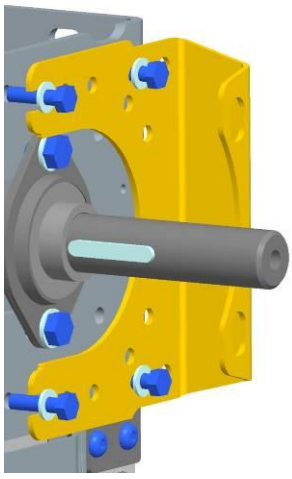


9. The bearing must be installed on the **drive side** (shaft with feather key), from **outside**. Lay the cables through the notch in the gusset plate.

10. The bearing must be installed on the **opposite side** (non drive side with short stub shaft) from **inside**.



300 plus
stainless steel

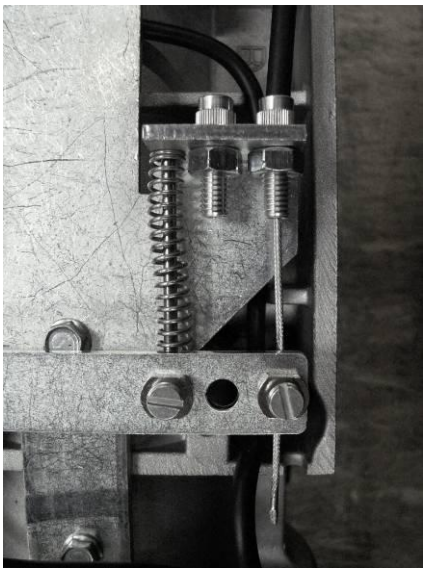


300 plus Alu



11. Affix the drive, when attaching the drive pay attention to the feather key connection. Stainless steel doors: Mount 2 washers between the gusset plate and the torque arm to avoid distortion. Screw the torque arm at the gusset plate.

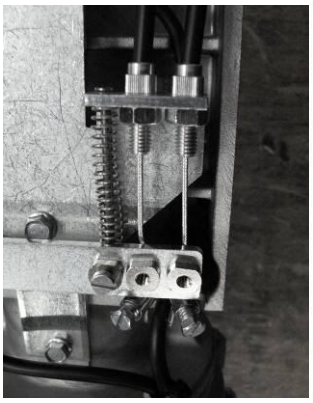
12. Drives with open circuit brake: Turn on power on drive before you roll off the curtain. **Brake is open without power!!**

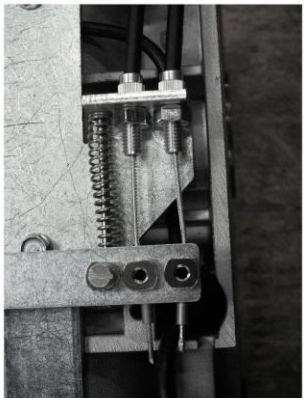


Design with fail safe brake and one release handle

Bowden cable clamped with screws

Bowden cable clamped with crimp sleeves

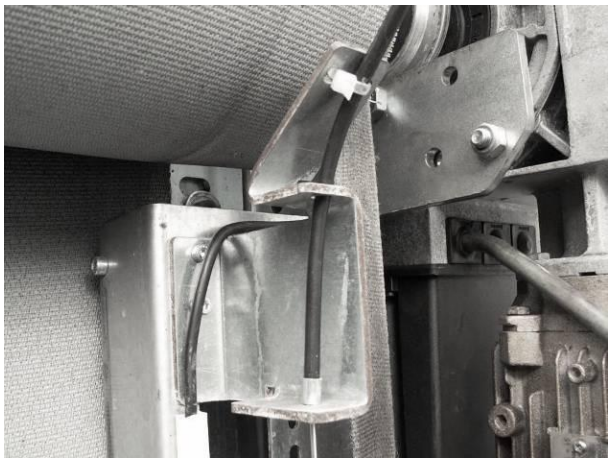




Design with fail safe brake and two release handle

13. Pull the Bowden cable through the sleeve and fix it with the screw at the brake lever of the drive. Pass the Bowden cable from the sleeve to the bracket in the side frame.

14. Pull the Bowden cables through the sleeves, do not fix it with the screw at the brake lever of the drive. Clamp the Bowden cable underneath the sleeve (with screws or crimp sleeves).



Doors with fail safe brake



15. Pass the Bowden cable at the side frame. Use a cable tie to fasten the Bowden cable at the bracket.

16. 300plusAlu: remove the screws from the cover profile. To open the side frame, turn the cover profile towards the frame profile and pull it out.



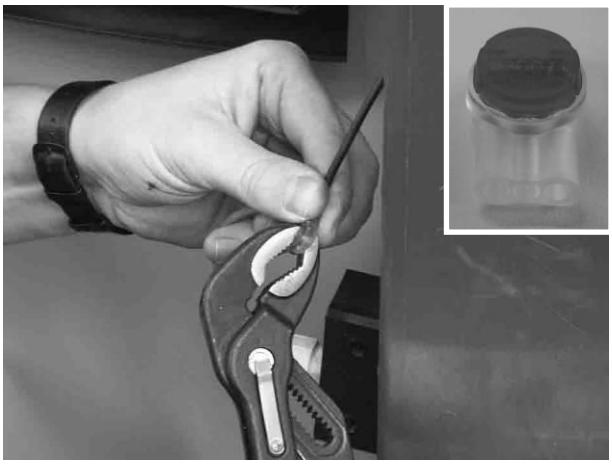
17. Lower the door curtain to work height. The drive brake must be released in order to turn the top roll by hand. (By operation of brake or disconnection of power).



18. For doors with stainless steel bottom beam the screws must be loosen before you slide on the bottom beam on the curtain. Slide the bottom profile on the reinforcement.



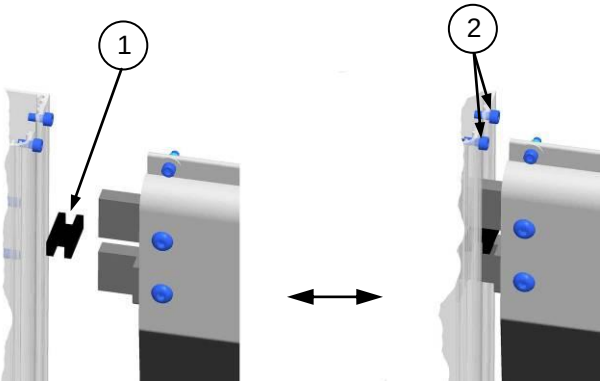
19. The side frame on the drive side is delivered with pre-installed cable chain and guide piece.
Doors with Crash-System are with a helix cable; these doors are without cable chain.

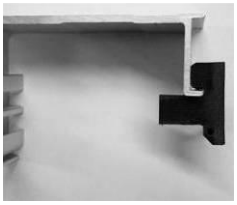


20. Install electrical connection between safety edge and cable chain (or helixcable).
Do **not** remove the wire insulation; press connector together with a pliers.
By 300plusAlu without Crash-System the connection is pluggable (M8 snap-in Plug)



Doors with curtain tension: See picture and text A1 in Appendix A at the end of the installation instructions.

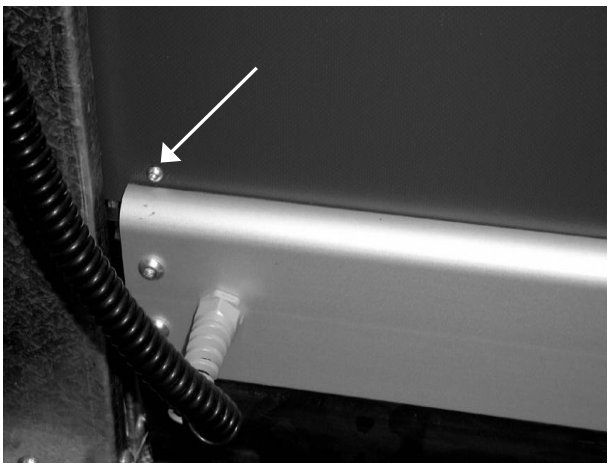


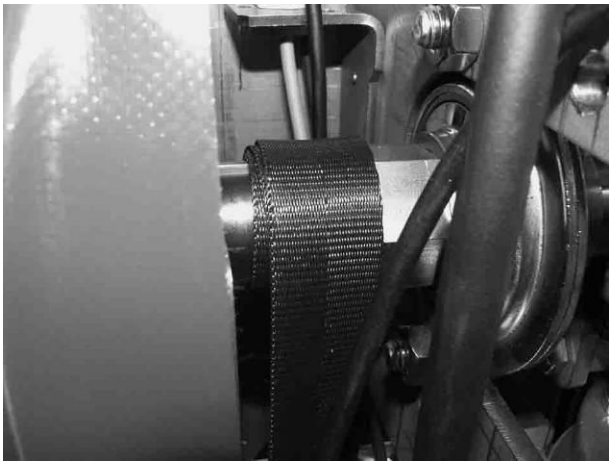


300plusAlu

21. In order to install the guide piece on the drive side, the guide-piece must be inserted in the bottom profile on the bearing side.
Align the bottom profile to the middle position of the curtain.

22. Place the H-piece **Pos. 1** in the guide of the side frame and push it into the slide unit groove. Screw the screws **Pos. 2** into the side frame (side frame top). By 300plusAlu two bell mouths are pre-installed.





23. Aluminium bottom beam: Drill holes into the door panel from the back side and fix the bottom profile with 2 screws M5x6.
Stainless steel bottom beam: Tighten the screws you have released before. (Picture 17).

24. Turn the top roll manually until the bottom profile is in the top position.
2-3 belt windings must remain on the shaft.





Left side shown



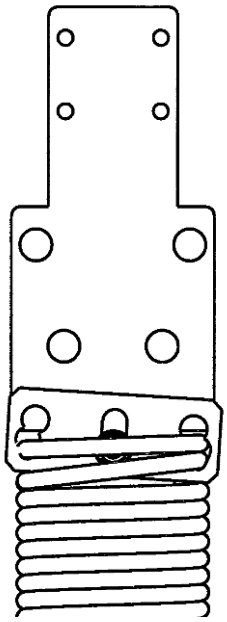
25. Description of routing the belt at the spring suspension top. Define belt length that the top end of the spring is at the marked position. Doors without curtain tension are not delivered with pulleys.

26. If doors are fitted with a curtain tension system then follow the instructions in **Appendix A** "Rope routing - curtain tension system" at the end of the installation instructions.









Shown: spring suspension top with cpl. assembled spring

27. Hook the spring into lower spring fixture. Check self opening of the door by operation of brake or release handle (fail safe brake, standard) or disconnection of power (open circuit brake).

28. The spring and the spring suspension top are cpl. assembled. In case of spring demount, the suspension could be turned to be taken out of the spring. Notice: The first turn is tight.



29. Stick on brush at side frame to reduce vibration and noises of springs, if necessary. Not with option sealed.



Doors with Crash-System

30. Fix the cable retainer onto the side part with 2 blind rivets. Tighten the plastic screws by hand.



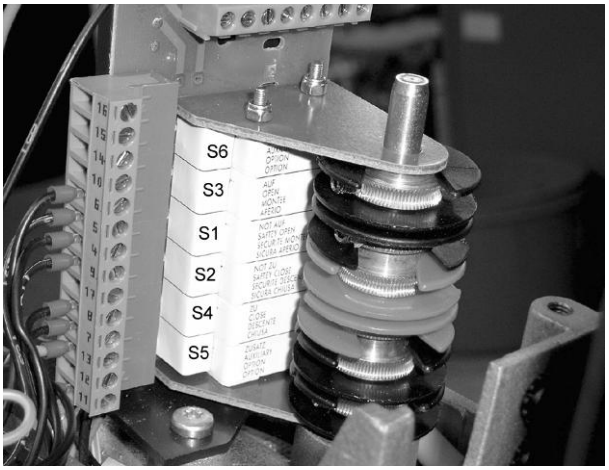
Doors with fail safe brake



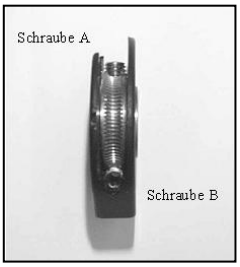


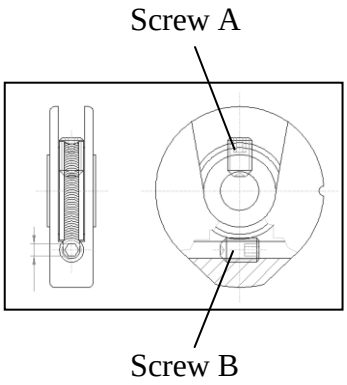
31. Hook in the spring in the hand lever before you hook in the spring cover on the drive side.

32. Hook in the spring cover and screw it on from outside. No screw in area of motor cover.
Turn the Aluminium cover profile towards the frame profile, snap the cover in, and screw it on.
Doors with fail safe brake: Operate the hand lever and adjust the Bowden cable if necessary.



S1 Emergency UP
S2 Emergency CLOSED
S3 UP
S4 CLOSED
S5 CUTOFF STAT. PHOTOCELL
S6 ADDITIONAL

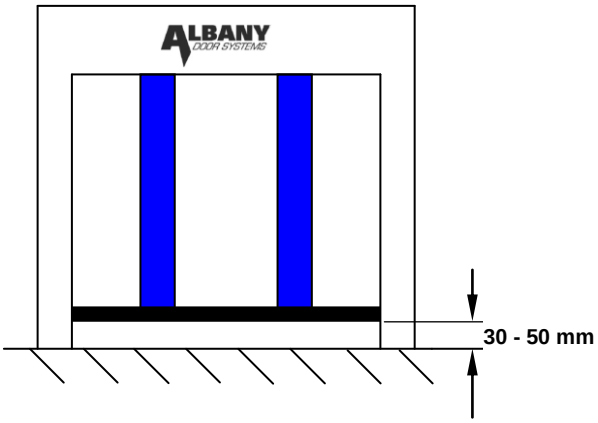


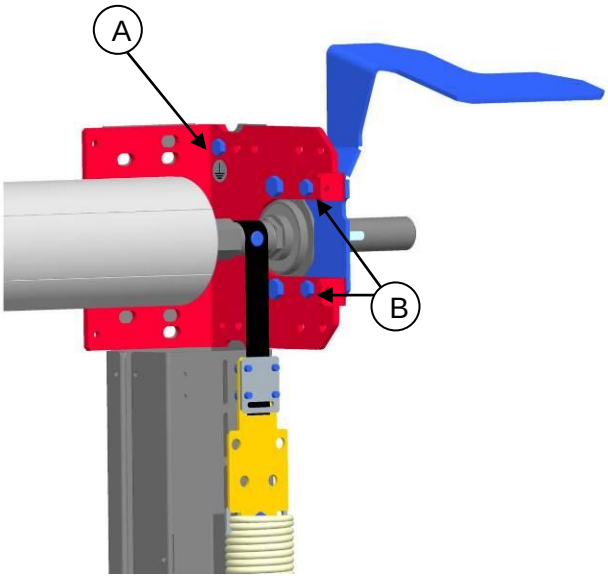


33. Connect electric power to drive. Set the cam limit switch in accordance with the end positions. **S4 and S5 are only used with ACS50. Wrong setting may lead to door damage!**

34. After the primary adjustment tighten **screw A** so that it cannot loosen (secure with paint). Adjust the end limits by turning **screw B**.

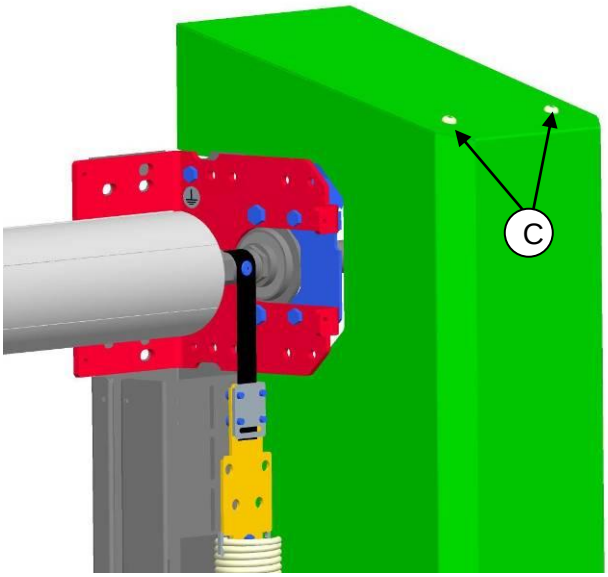
Program lower limit (S4)!
Only with ACS 50





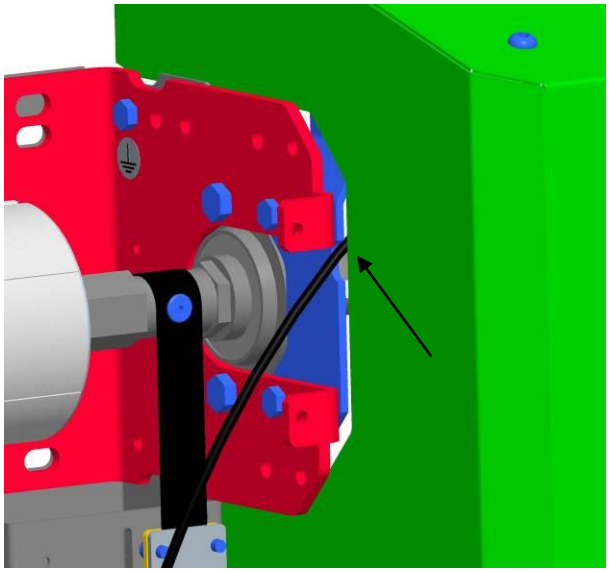
35. Set the cam limit switch **S4** so that it is activated by an opening height of 30 - 50 mm. At doors with MCC the closed position and the cutoff of the stat. photocell is set at the control.

36. Connection for potential equalisation feasible **(A)**!
Screw the bracket for the motor cover at the head plate **(B)**.



37. Slide the motor cover on the bracket for the motor cover and screw it at the bracket **(C).**

Doors with fail safe brake and motor cover

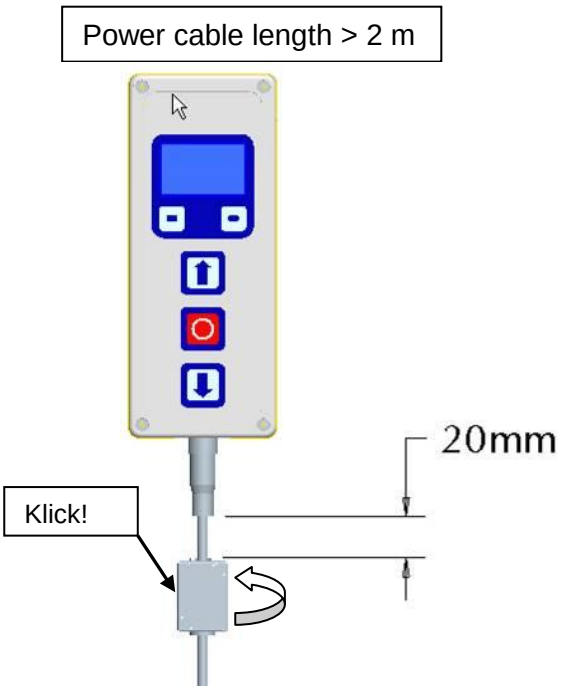


38. Direct the Bowde as shown. (Not for doors with open circuit brake)





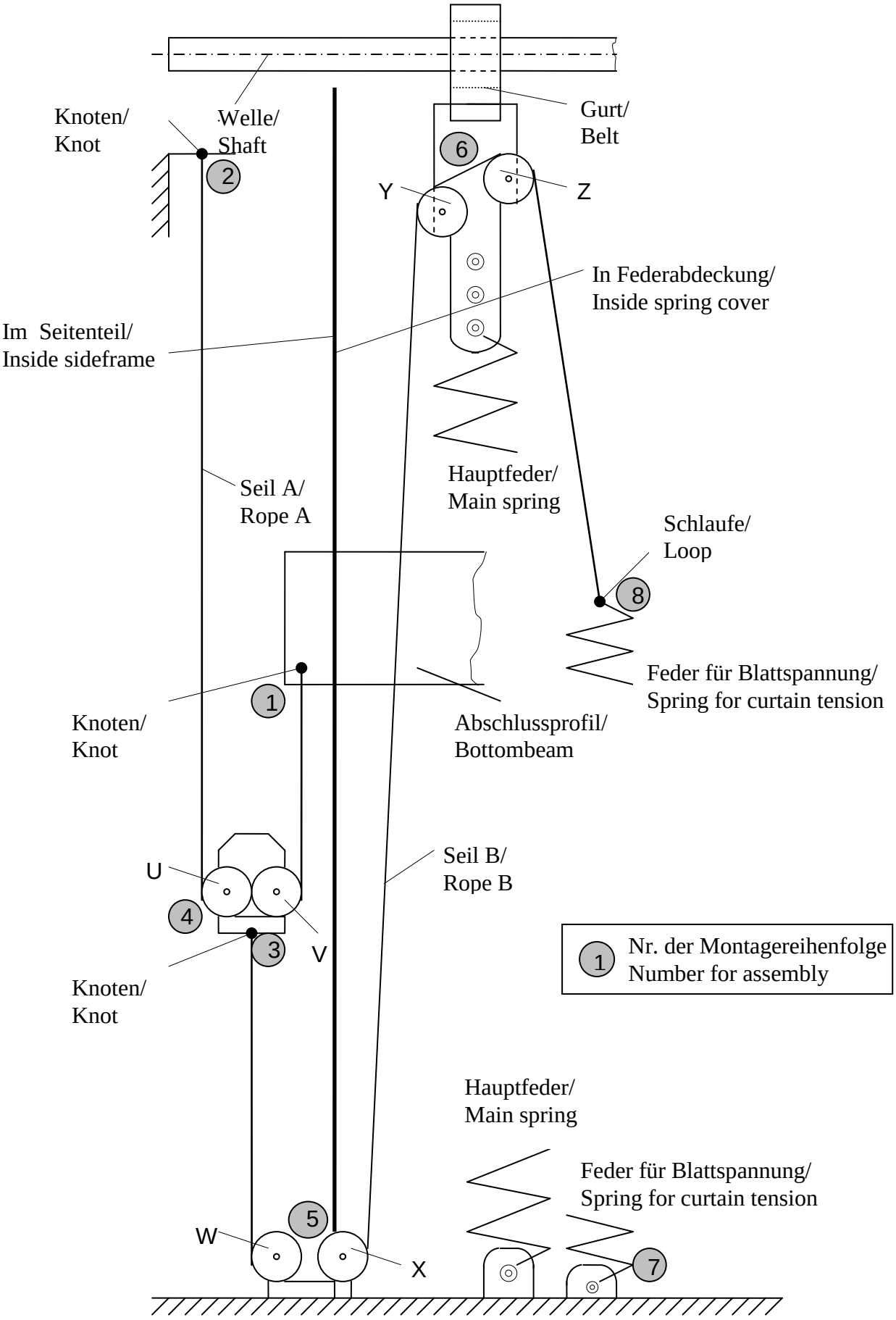
39. Anchor top roll cover with M8 and sidecover (300plusAlu) with M10 screws to the head plate.

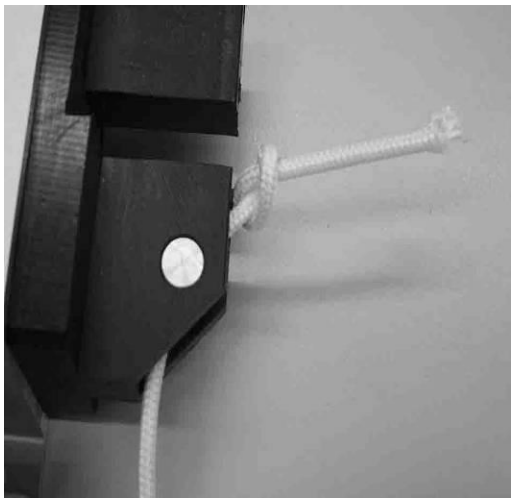


40. After testing the safety fixtures and executing a trial run, the installation procedure is concluded.

The door is dismantled in the reverse sequence of the installation procedure. The door's packaging can be returned to the manufacturer. Dispose of defective door components in an environmentally responsible manner. Defective electronic components must be disposed of as special waste.	

Seilführung Blattspannung – linke Seite dargestellt/
Rope for curtain tension – left side shown





A1. Push the bolt into the guide-piece, pass **rope A** through, and tie a knot.
300plusSteel/StainlessSteel: **rope A** (L=DH+600)
300plusAluminum: **rope A** (L=DH+100)
Close the door complete!





A2. **300plusSteel/StainlessSteel:** Guide **rope A** through the bore in the angle bracket (on top of side frame) and tie a knot. **300plusAlu:** Loose the screw from the frame profile, pass **rope A** through channel, tie a knot, and tight the screw. 2 washers necessary!



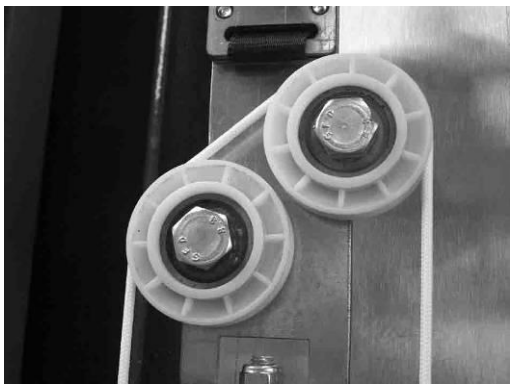


A3. **300plusSteel/StainlessSteel:** Disassemble **rollers (U** and **V)** to pull **rope A** through the roll holder. Connect **rope B** with a knot.
300plusAlu: Instead of two Rollers (U and V) => just one roller; otherwise as by 300plusSteel/ StainlessSteel.
Pull on **rope B** to apply tensile load to **rope A**!





A4. Guide **rope B** under the two lower **rollers W** and **X** from the inside to the outside.



Drawn spring bracket side frame left, right side is mirror inverted

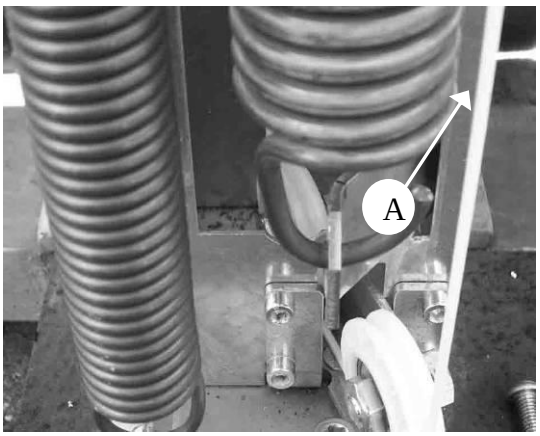


A5. Outside side frame:
Guide **rope B** over the two **rollers Y** and **Z** on the upper spring fixture (hangs on the belt).

A6. Make a loop in **rope B** and hook in the spring. When the door is **completely closed** the roll holder should not hit the **roller W; <u>Distance about 20mm</u>.** Make a 2nd knot in rope A, if necessary (see A2). Tense the spring of curtain tension up to the label.







A7. Hook in the spring of curtain tension at the bottom (small spring).

A8. Position **A** shows the area, where the gap between spring and rope must be appr. 3mm. If there is no gap => bend the spring bracket with a pliers.